

# Mario Avolio

AI & CLOUD ENGINEER · GENERATIVE & AGENTIC AI

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## Summary

AI & Cloud Engineer focused on production-grade generative AI: LLM applications, agentic workflows, and RAG pipelines, served as containerized microservices on the cloud. Research background in computer vision and model optimization, with a Best Paper Award at ICIAP 2025. I build reliable, evaluable systems that move from prototype to production.

## Experience

### Machine Learning Reply

Milan, Italy

MACHINE LEARNING ENGINEER

Jan. 2026 - Present

- Build agentic workflows, RAG pipelines, and LLM-powered automation for enterprise use cases, from prototyping through deployment.
- Deploy and operate generative-AI services on Azure cloud infrastructure with containerized microservices and CI/CD.
- Apply evaluation and observability practices to keep AI systems reliable once exposed to real users and data.

### Imaging and Vision Laboratory (IVL), University of Milano-Bicocca

Milan, Italy

RESEARCH FELLOW

Jan. 2025 - Dec. 2025

- Built a large-scale synthetic Side-Scan Sonar dataset of 20,000+ labeled images for underwater mine detection, beyond the scale of existing public datasets.
- Applied and optimized super-resolution models (SPAN, DRCT) to recover detection performance comparable to high-resolution data.
- Trained lightweight YOLOv11n/s detectors for robust detection in low-visibility conditions; work received a Best Paper Award at ICIAP 2025.

## Skills

### Generative AI & LLMs

LLM applications, RAG, multi-agent / agentic workflows, tool-calling, prompt & evaluation design, fine-tuning (LoRA/PEFT), vector databases

### Frameworks & Tooling

LangChain, OpenAI Agents SDK, Hugging Face, Ollama, ChromaDB, Pydantic

### Deep Learning & CV

PyTorch, self-supervised learning (DINOv2), super-resolution, model optimization (quantization, pruning, distillation)

### Cloud & MLOps

Azure, Docker, FastAPI, CI/CD (GitHub Actions), uv, pytest, model serving & observability, Linux

### Programming

Python, SQL, Bash, Java, C++

### Languages

Italian (native), English (fluent)

## Projects

### RAG · Multi-agent · Cloud

AI MICROSERVICES HUB

[github.com/MarioAvolio/Portfolio](https://github.com/MarioAvolio/Portfolio)

- A FastAPI gateway fronting independent generative-AI microservices: a profile RAG assistant (LangChain + ChromaDB), multi-agent research (OpenAI Agents SDK), and text intelligence, with pluggable LLM providers (OpenAI, Gemini, Hugging Face, local Ollama). Each containerized with its own tests and CI; Python, uv, Docker Compose.

## Education

### University of Milano Bicocca

Milan (MI), Italy

M.S. IN COMPUTER SCIENCE (GRADE: 110/110 CUM LAUDE)

Sept. 2021 - March 2024

- **Thesis:** Self-Supervised Learning and Model Adaptation for Facial Attribute Classification (DINOv2, LoRA, PEViT).

### University of Calabria

Arcavacata (CS), Italy

B.S. IN COMPUTER SCIENCE (GRADE: 108/110)

Sept. 2018 - Sept. 2021